

LESSON 12.2

FINDING SURFACE AREA OF RIGHT CIRCULAR CYLINDERS



LESSON INTRODUCTION

MA.7.GR.2.1 Given a mathematical or real-world context, find the surface area of a right circular cylinder using the figure's net.

MA.7.GR.2.2 Solve real-world problems involving surface area of right circular cylinders.

LEARNING TARGETS

- I can draw the net of a right circular cylinder.
- I can find the surface area of a right circular cylinder.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.GR.4.6 Solve mathematical and real-world problems involving the surface area of three-dimensional figures limited to cylinders, pyramids, prisms, cones, and spheres.

ESSENTIAL QUESTIONS

- What do you need in order to calculate the surface area?
- How do you calculate the surface area when given its diameter and its height?
- How is the sum of the parts of the net equal to the surface area of the cylinder?

LESSON NARRATIVE

The focus of this lesson is to find the surface area of a right circular cylinder without the figure's net when given the measure of the radius or diameter and the height of the cylinder. Students will work through finding the surface area by breaking down the cylinder into two circular bases and a rectangle for the lateral area. Students will see problems in real-world context.

COMPONENT SUMMARY

12.2.1 WARM-UP (~5 MIN.)

Career Video

12.2.3 GUIDED PRACTICE (10 MIN.)

Practice problems on calculating surface area

12.2.5 WRAP-UP (~5 MIN.)

Self-Assessment

12.2.2 GUIDED INSTRUCTION (20 MIN.)

Finding surface area without using the nets

12.2.4 YOUR TURN! (~5 MIN)

Independent practice on finding surface area

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Right circular
- Cylinder
- Lateral area

MATERIALS

- Career Profile Video
- Calculators
- (Optional) Chart Paper
- (Optional) Markers

ADDITIONAL PREPARATION

- 12.2.1 Warm-Up contains a 3-minute video to accompany the question prompts.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may use the diameter for radius when finding surface area.
- Students may struggle calculating the surface area without the image's net.
- Students may struggle leaving answers in terms of π , either not recognizing the equivalence of the answer when multiplied by π .

THINGS TO CONSIDER

- Correct and consistent use of vocabulary will help to build students' competence. This will be particularly helpful for ELL students.
- If students continue to struggle, encourage students to draw and label the net of the real-world problems.
- Consider making questions in 12.2.2 Guided Instruction no-calculator questions and allowing a calculator in 12.2.3 Guided Practice.
- If time is an issue, then consider assigning 12.2.4 Your Turn! for homework.

LITERACY AND LANGUAGE ROUTINES

Compare and Connect

This lesson provides students the opportunity to find the surface area of a right circular cylinder without using the net of the figure. In 12.2.3 Guided Practice, encourage students to compare and connect finding the surface area with and without using the figure's net. Allow students to review examples from Lesson 1 so they have a strong understanding of how to calculate the surface area.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.3.1 Mathematical Fluency

Mathematical Fluency covers being precise while carrying out calculations. By providing students with multiple opportunities to execute procedures and perform calculations, this lesson allows students to accurately find the surface area of right circular cylinders.

DIFFERENTIATE INSTRUCTION

12.2.2 Guided Instruction provides both visual and auditory entry points for learners to engage with the content. The table provides students with a visual on finding the surface area by first finding the area of each part, then finding the sum of all the parts. Consider providing students opportunity and time to sketch alternative representations that make sense to them, like utilizing and labeling manipulatives in Lesson 1 and the toilet paper roll.

Additional differentiation opportunities are denoted throughout the lesson guide.

12.2.1 WARM-UP: MANUFACTURING ENGINEER

QUESTIONING

Consider inviting an engineer to join the class virtually or in-person. Additionally, consider having students return to this lesson later in the year for project-based learning around this Career Profile.



12.2.1 Warm-Up: Manufacturing Engineer

Joanel Johnson, a senior manufacturing engineer, mentions that one of his responsibilities is to interpret and explain, in layman's terms (common language), the drawings of the design engineers to the technicians he works with.

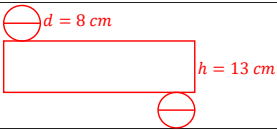
1. Describe a time when you had to interpret for yourself or explain a math problem to a partner. What did you do to ensure that you or your partner understood the problem?

Answers will vary. I ensured that I understood the problem by making sure that I asked for help if I could not interpret it myself. I ensured that my partner understood the problem by not giving up even when it seemed hard to explain it to them.



12.2.2 Guided Instruction: Finding Surface Area of Right Circular Cylinders

1. Assume that a can of soda is in the shape of a right circular cylinder.
- a. Complete the table to find the surface area of a can of soda whose diameter is 8 centimeters (cm.) with a height of 13 cm. Use $\frac{22}{7}$ for π . Round your answer to the nearest thousandth.

Draw a net and label its dimensions. (Include units.)	
Determine the area of one circular base. (Include units.)	50.286 cm ²
Find the sum of the areas of both circular bases. (Include units.)	100.572 cm ²
Find the lateral area. (Include units.)	326.857 cm ²
Find the sum of the areas for all the parts. (Include units.)	Sum of the Areas of the Circular Bases + Lateral Area ≈ 427.429 cm ²

- b. What is the surface area of a can of soda whose diameter is 8 centimeters (cm.) with a height of 13 cm?
- ≈ 427.429 cm²

12.2.2 GUIDED INSTRUCTION: FINDING THE SURFACE AREA OF RIGHT CIRCULAR CYLINDERS

TEACHER MOVES

This component is designed to scaffold student discovery of how to find the surface area. The lesson does not rely on the formula, but on using student understanding of nets, area of circles, and area of rectangles to build student understanding of how to find the surface area. Benchmark Clarification 1 states that “the expectation is not to memorize the surface area formula for a right circular cylinder.” Use the scaffolding in the table to help build student understanding without a reliance on the formula. Help build students’ fluency with rational numbers by having them find the areas without a calculator.

TEACHER MOVES

Think-Pair-Share

- This strategy can be used on questions 1a and 1b.
- Students should work on each question individually and record answers.
 - Students will pair with their partner or group members to discuss solutions and provide constructive critiques.
 - The teacher will select a few groups to share their solutions and methods.

12.2.2 GUIDED INSTRUCTION: FINDING THE SURFACE AREA OF RIGHT CIRCULAR CYLINDERS (CONTINUED)

TEACHER MOVES

Consider having a class discussion on what it means to leave an answer in terms of π .

DIFFERENTIATE INSTRUCTION

Consider providing students with the following number/sentence frame to fill in to help them find the surface area of the right circular cylinder. Use colors if available.

$$2(\text{_____}) + \text{_____} = \text{_____}$$

$$2(\text{area of circular base}) + \text{lateral area} = \text{surface area}$$

TEACHER MOVES

Notice and Wonder

Consider providing students with the opportunity to write down things they notice and wonder about Charlie's work. This will allow students the chance to make observations before answering question 3.

- c. Your partner is absent from class today. Write a summary describing to your partner how to find the surface area of a right circular cylinder when given the cylinder's height and diameter.

Answers will vary. To find the surface area, first find the radius of the circular base. Then, find the combined area of the circular bases. Next, find the circumference of one circular base and multiply it by the height of the can. Finally, add the product of the lateral area, to the area of the circular bases.

2. Evelyn stores all her gift wrap paper in a plastic storage bag that is shaped like a right cylinder. The storage bag is 41 inches (in.) tall, and its radius measures 5.6 in. What is the surface area of Evelyn's storage bag? Leave your answer in terms of π and include units.

$$521.92\pi \text{ in}^2$$

3. Charlie calculated the surface area of a mailing tube whose diameter is 5 inches (in.) and height is 26 in. His calculation is shown. Work with your partner to determine if Charlie is correct. If not, identify his error(s) and correct his work.



$$\begin{aligned} &(\pi \cdot 5^2) + 2 \cdot \pi \cdot 5 \cdot 26 \\ &25\pi + 10\pi \cdot 26 \\ &25\pi + 260\pi \\ &285\pi \end{aligned}$$

Charlie is not correct. In the first line, Charlie should have used the radius, 2.5, instead of the diameter. In second line, Charlie should have added the area of both circular bases instead of one area. In the last line, Charlie should have found the surface area to be $142.5\pi \text{ in}^2$.



12.2.3 Guided Practice: Finding Surface Area of a Right Circular Cylinder

1. There are three different types of bass drums: tom-toms, floor toms, and the kick drum. Each drum is the shape of a right circular cylinder. Suppose you intend to place each drum individually in a heavy-duty nylon bag when traveling locally. Find the amount of nylon needed for each drum bag. Use 3.14 for π and include units in your answer.



Bass Drum	Work	Amount of Nylon Needed
Tom-Tom $d = 7$ in. $h = 10$ in.	$[(3.5^2)(3.14)(2)] + [(10)(7)(3.14)] =$	296.73 in^2
Floor Tom $d = 15$ in. $h = 16$ in.	$[(7.5^2)(3.14)(2)] + [(16)(15)(3.14)] =$	1106.85 in^2
Kick Drum $r = 8$ in. $h = 22$ in.	$[(8^2)(3.14)(2)] + [(22)(16)(3.14)] =$	1507.2 in^2

12.2.3 GUIDED PRACTICE: FINDING SURFACE AREA OF A RIGHT CIRCULAR CYLINDER

DIFFERENTIATE INSTRUCTION

For students who are struggling, encourage them to draw the figure’s net to use a visual. Have students find the area of each part, then calculate the sum of all the parts.

QUESTIONING

Consider challenging students who finish early by finding the lateral area of each bass drum.

12.2.4 YOUR TURN!

TEACHER MOVES

Some students may struggle without the scaffolding. Consider having students draw the net or set up a table like in question 1 in 12.2.2 Guided Instruction.



12.2.4 Your Turn!

1. A small business on Etsy sells confetti cannons for buyers to celebrate special occasions. The confetti cannons come in an enclosed tube in the shape of a right circular cylinder. Each cannon is 6 inches (in.) tall and 1.5 in. wide. How much material is used to make the enclosed tube of each cannon? Leave your answer in terms of π and include units.

$10.125\pi \text{ in}^2$



12.2.5 Wrap-Up: Reflection

1. Take a **SELFIE** by checking the box that best matches what you did today.
Answers will vary.

Showed my work	☐	☐
Explained my answers	☐	☐
Lots of math vocabulary used	☐	☐
Found multiple solutions	☐	☐
I persevered through the problem	☐	☐
Eliminated careless errors	☐	☐

12.2.5 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

Ask students to elaborate on their reflection. Provide sentence stems such as:

- I did not _____ today because_____.
- I struggled with _____ today because_____.